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RTP (Real-time Transport Protocol)

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Course Name: **Data Communication**

Submitted to:

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Outline

01 Introduction and research problem

02 Related work and research gap

03 How RTP works (methodology)

04 Protocol features and comparison

05 Conclusion

Introduction and Research Problem

Live audio and video need timely delivery, but TCP and plain UDP do not provide it.

01 Why it matters: VoIP, video calls and live streaming are used every day

02 Gap: TCP adds delay on loss; UDP has no timing or ordering information

03 Objective: understand how RTP and RTCP solve this over UDP

Related Work and Research Gap

Study / approach	Key idea	Relevant limitation
TCP (RFC 9293)	Reliable, ordered byte stream	Retransmission adds delay to live media
Plain UDP	Fast, connectionless datagrams	No timestamps, ordering or payload type
RTP (RFC 3550)	Timing and sequencing on top of UDP	No QoS guarantee, no built-in encryption

Research gap

Real-time media needs timing and loss detection without TCP's delay; RTP and RTCP provide this.

Methodology

RTP and RTCP over UDP/IP

RTP fixed header: 12 bytes, 32 bits per row (field sizes in bits)

V 2	P 1	X 1	CC 4	M 1	PT 7	Sequence no. 16
Timestamp (32)						
SSRC: synchronisation source ID (32)						
CSRC list: 0 to 15 items (optional)						

V version, P padding, X extension, CC CSRC count, M marker, PT payload type

01 Packetise

Encode the media, split it into packets and add the RTP header

02 Send over UDP

Each packet carries a sequence number and a timestamp

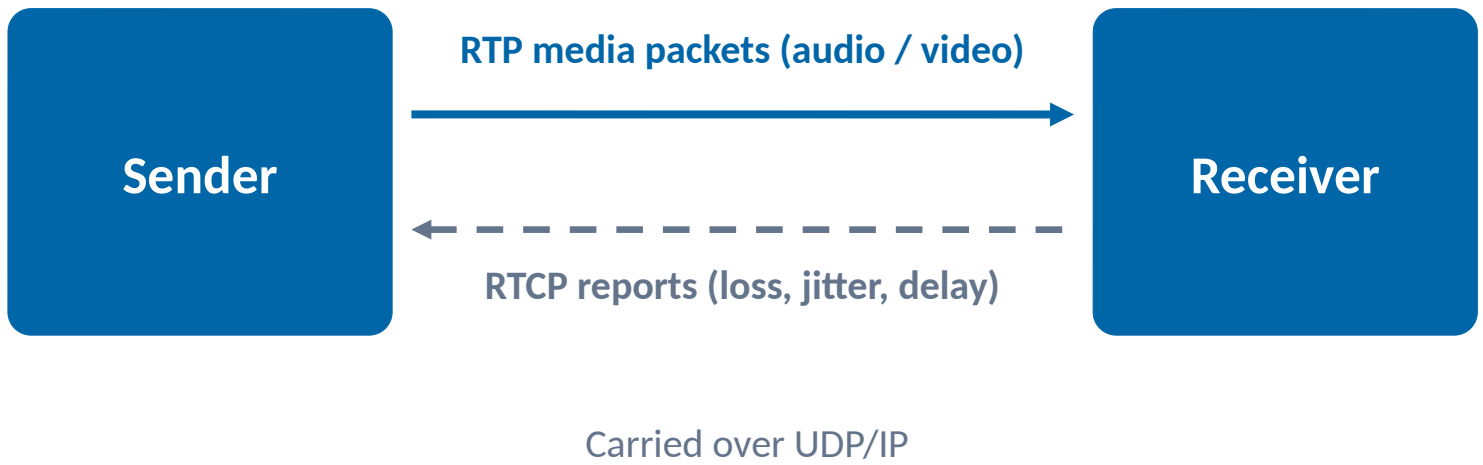
03 Play out and report

Receiver reorders, absorbs jitter and sends RTCP reports

Protocol Overview

Transport / ports	Usually over UDP; RTP on an even port, RTCP on the next odd port
Packet format	12-byte fixed header plus optional CSRC list, extension and payload
Timing / clock	Timestamp follows the media clock, e.g. 8 kHz audio or 90 kHz video
Standards / security	RFC 3550 and RFC 3551 define RTP; SRTP (RFC 3711) adds encryption

RTP and RTCP in Action



Main finding

RTP delivers media with sequence numbers and timestamps.

RTCP reports loss, jitter and delay so the sender can adapt.

Figure: RTP media flow (top) and RTCP feedback (bottom) between sender and receiver.

Comparison with TCP and UDP

Method	Timing info	Loss detection	Latency
TCP	No	Yes, retransmits	Higher on loss
UDP	No	No	Low
RTP over UDP	**Yes (timestamp)**	**Yes (sequence no.)**	**Low**

RTP keeps UDP's low latency and adds the timing and ordering that live media needs.

Conclusions and Future work

RTP with RTCP makes real-time audio and video practical over IP networks.

01 Contribution: timing, sequencing and payload identification for live media

02 Limitation: no QoS guarantee and no built-in encryption (SRTP adds it)

03 Next step: study WebRTC and SRTP in modern calls and streaming

Selected References

[1] H. Schulzrinne, S. Casner, R. Frederick and V. Jacobson, “RTP: A Transport Protocol for Real-Time Applications,” RFC 3550, IETF, 2003.

[2] H. Schulzrinne and S. Casner, “RTP Profile for Audio and Video Conferences with Minimal Control,” RFC 3551, IETF, 2003.

[3] M. Baugher et al., “The Secure Real-time Transport Protocol (SRTP),” RFC 3711, IETF, 2004.

[4] H. Alvestrand, “Overview: Real-Time Protocols for Browser-Based Applications,” RFC 8825, IETF, 2021.

Thank you

Questions and Discussion

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